

AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media

A comparative Beamer presentation for diploma defence

Kasymkhanov Timur Askarov Insar Kaidrakhmetova Dinara Nurshanov Dias

International Information Technology University

Bachelor's diploma defence, 2026

Research adviser: **Bakhyt N. Alipova** (PhD, Applied Mathematics)

Motivation and Research Problem

Motivation

Temperature changes in rocks can affect stress, deformation, and wave propagation. This is relevant to geothermal systems, underground engineering, and geophysical interpretation.

Research problem

Estimate directional thermoelastic response using AI-assisted surrogate models under simplified effective-medium assumptions.

Scope: research prototype for source–probe prediction, not field-validated industrial simulation.



Aim, Tasks, and Scope

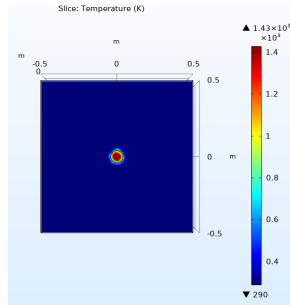
Aim

Develop and justify an AI-guided framework for directional prediction of thermoelastic wave propagation in geological media.

1. Define the physical and geological basis.
2. Formulate a simplified 2D thermoelastic prediction task.
3. Prepare COMSOL-derived synthetic data.
4. Compare PINN, FNO, MeshGraphNet, and Transformer-style routes.
5. Interpret results, limitations, and future improvements.

Object: thermoelastic wave propagation in geological media.

Subject: AI-assisted source–probe directional prediction.



Controlled 2D thermoelastic field data support comparison.

Thermoelastic Coupling Logic

1. **Thermal excitation:** a local source changes the temperature field.
2. **Thermal expansion:** temperature change produces local strain.
3. **Stress formation:** constrained expansion creates thermal stress.
4. **Mechanical response:** stress gradients drive displacement.

$$T \longrightarrow \varepsilon_{\text{th}} \longrightarrow \sigma \longrightarrow u$$

Physical meaning

Temperature change causes thermal strain; thermal strain contributes to stress; stress affects displacement and wave propagation.

T – temperature [K]; ε_{th} – thermal strain [dimensionless]; σ – stress [Pa]; u – displacement [m].

Geological Media and Effective Parameters

Rock	ρ kg/m ³	V_P m/s	k W/(m K)	C_p J/(kg K)
Granite	2650	5850	2.50	850
Basalt	3000	6550	1.95	1300
Sandstone	2725	5000	2.25	1000
Limestone	2675	5400	2.55	1050

Effective scalar parameters support comparison under a locally homogeneous isotropic approximation, not field calibration.



Public-domain limestone texture, Wikimedia Commons / Titus Tschardtke.

Mechanical Rock Parameters

Material	Type	ρ kg/m ³	E GPa	ν –	μ GPa	K GPa	V_P/V_S m/s
Granite	intrusive	2650	76.28	0.245	30.63	49.84	5850/3400
Granodiorite	intrusive	2795	85.91	0.255	34.24	58.35	6100/3500
Basalt	volcanic	3000	103.96	0.266	41.07	73.95	6550/3700
Diabase	hypabyssal	2975	98.22	0.274	38.56	72.36	6450/3600
Gabbro	mafic	3075	114.27	0.287	44.40	89.33	6950/3800
Sandstone	sedimentary	2725	55.75	0.259	22.13	38.61	5000/2850
Limestone	carbonate	2675	61.48	0.277	24.08	45.90	5400/3000
Marble	metamorphic	2725	82.41	0.270	32.43	59.82	6150/3450
Schist	foliated	2800	68.20	0.267	26.91	48.82	5500/3100
Quartzite	quartz-rich	2650	83.50	0.250	33.40	55.70	6150/3550

Elastic constants are effective midpoint values derived under isotropic assumptions for model-ready comparison.

Thermal and Porosity Parameters

Material	α $10^{-6}/\text{K}$	k $\text{W}/(\text{m K})$	C_p $\text{J}/(\text{kg K})$	C $\text{MJ}/(\text{m}^3 \text{K})$	γ MPa/K	Porosity %
Granite	7.9	2.50	850	2.25	1.18	total 45.5
Granodiorite	—	2.40	—	—	—	—
Basalt	—	1.95	1300	3.90	—	total 19.0
Diabase	—	2.50	850	2.53	—	—
Gabbro	—	2.30	1000	3.08	—	total 43.5
Sandstone	11.6	2.25	1000	2.73	1.34	total 31.5; eff. 26.5
Limestone	8.0	2.55	1050	2.81	1.10	total 31.5; eff. 18.0
Marble	9.8	2.80	850	2.32	1.76	—
Schist	—	2.75	1150	3.22	—	total 26.5; eff. 27.5
Quartzite	—	5.10	1000	2.65	—	—

Missing thermal expansion or coupling values are shown explicitly as dashes rather than inferred.

Rock Properties Controlling Propagation

Mechanical controls

- ▶ Density controls inertia.
- ▶ Elastic stiffness controls resistance to deformation.
- ▶ Porosity, cracks, and saturation change effective stiffness.

Thermal controls

- ▶ Conductivity controls heat transfer.
- ▶ Heat capacity controls thermal storage.
- ▶ Thermal expansion converts temperature change into strain.

Directional behavior can appear because bedding, foliation, aligned pores, fractures, and stress state make real rocks structurally ordered.

Elastic Waves in Geological Media

Compressional (P) waves

Compressional waves. Particle motion is mainly parallel to the propagation direction and involves volume change.

Shear (S) waves

Shear waves. Particle motion is mainly transverse to the direction of propagation and depends on shear rigidity.

$$V_P = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}, \quad V_S = \sqrt{\frac{G}{\rho}}$$

V_P, V_S – compressional and shear wave velocities [m/s]; K – bulk modulus [Pa]; G – shear modulus [Pa]; ρ – density [kg/m³]. These relations show that stiffness increases wave velocity, while density increases inertia.

Heat Conduction and Thermal Expansion

Fourier heat flux and heat equation

$$q = -k\nabla T, \quad \rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k\nabla T) + Q$$

q – heat flux [W/m²]; k – thermal conductivity [W/(m K)]; T – temperature [K]; ρ – density [kg/m³]; C_p – specific heat capacity [J/(kg K)]; t – time [s]; Q – volumetric heat source [W/m³].

Temperature perturbation and thermal strain

$$\theta = T - T_0, \quad \varepsilon_{\text{th}} = \alpha \Delta T$$

$\theta, \Delta T$ – temperature change [K]; T_0 – reference temperature [K]; ε_{th} – thermal strain [dimensionless]; α – linear thermal expansion [1/K].

Key Physical Parameters

Thermal diffusivity relation

$$\alpha_T = \frac{k}{\rho C_p}$$

- ▶ α_T is thermal diffusivity [m^2/s].
- ▶ k is thermal conductivity [$\text{W}/(\text{m K})$].
- ▶ ρ is density [kg/m^3], and C_p is heat capacity [$\text{J}/(\text{kg K})$].
- ▶ Larger α_T means faster spreading of a temperature disturbance.

Thermoelastic Coupling

Thermoelastic extension of Hooke's law

$$\theta = T - T_0$$
$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij} - \gamma \delta_{ij} \theta, \quad \gamma = 3K\alpha$$

Equation of motion

$$\rho \frac{\partial^2 u_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial x_j} + f_i$$

θ – temperature perturbation [K]; T, T_0 – current and reference temperatures [K]; σ_{ij} – stress tensor [Pa]; ε_{ij} – strain tensor [dimensionless]; ε_{kk} – volumetric strain [dimensionless]; λ, μ, K – Lamé and bulk moduli [Pa]; δ_{ij} – Kronecker delta [dimensionless]; α – thermal expansion [1/K]; γ – thermoelastic coupling coefficient [Pa/K]; u_i – displacement [m]; x_j – coordinate [m]; f_i – body force density [N/m³].

The relation states that stress depends on elastic deformation and temperature-induced expansion; this is why thermal and elastic fields are coupled in the model.

Mathematical Scope and Primary Fields

Simplified model scope

- ▶ 2D continuum: $\Omega \subset \mathbb{R}^2, (x, t) \in \Omega \times [0, T_f]$.
- ▶ In-plane displacement: $u(x, t) = [u, v]^T$.
- ▶ Small deformation and linear isotropic thermoelasticity.
- ▶ Pores, fractures, fluids, plasticity, damage, and full anisotropy are not resolved explicitly.

Field	Meaning	Role
$T(x, t)$	Temperature	Thermal state
$\theta(x, t)$	Perturbation	Thermal loading
(u, v)	Displacement	Probe response
ε_{ij}	Strain	Deformation
σ_{ij}	Stress	Internal force
$d, \hat{d}, \phi$	Direction	Source–probe query

These fields connect the physical formulation with the prediction methodology used in the comparative prototype.

These assumptions keep the task suitable for comparative AI-assisted directional prediction and make the results interpretable within the scope of a bachelor thesis. Here Ω is the model domain [m²], x is position [m], t is time [s], T_f is final time [s], and u, v are displacement components [m].

Directional Prediction Task



A source point and a probe point define the directional query.

$$d = x_p - x_s, \quad L = \|d\|, \quad \hat{d} = \frac{d}{\|d\|}$$

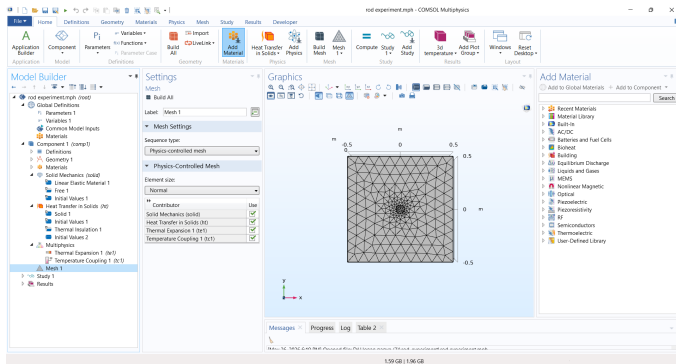
$$\phi = \text{atan2}(d_y, d_x), \quad \mathcal{F} : X \rightarrow \hat{Y}$$

x_s, x_p – source and probe positions [m]; d – direction vector [m]; L – distance [m]; $\hat{d}$ – unit direction [dimensionless]; ϕ – azimuth [rad]; $\mathcal{F}$ maps input features X to predicted outputs $\hat{Y}$.

Prediction contract

Input: material parameters, geometry, direction, and time. Output: estimated temperature, displacement, directional response, and travel time.

COMSOL-Based Synthetic Dataset



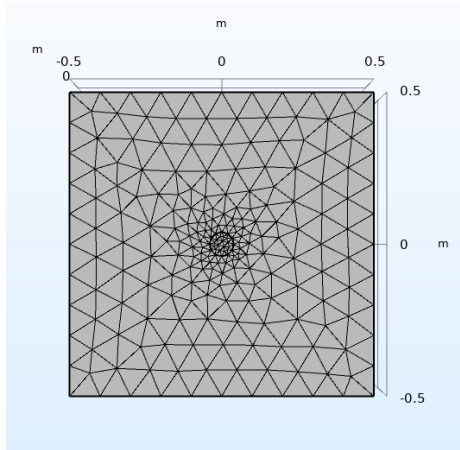
COMSOL simulation

The synthetic dataset was generated in COMSOL Multiphysics using a coupled thermoelastic model.

- ▶ 2D plane-strain geological domain
- ▶ Domain size: $1 \text{ m} \times 1 \text{ m}$
- ▶ Initial temperature: 293.15 K
- ▶ Source temperature: 1500 K
- ▶ Exported fields: T , (u, v) , stress, strain, velocity

The exported COMSOL fields are used as synthetic reference data for comparing neural surrogate models.

COMSOL 2D Model and Mesh



Model domain

The numerical experiment represents a two-dimensional section of a geological medium with an internal heated source.

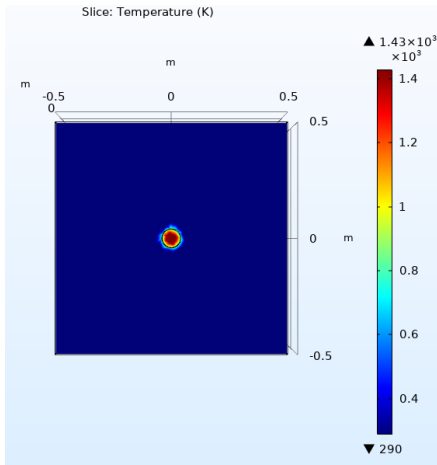
Mesh refinement

The mesh is refined near the source, where the strongest temperature, displacement, and stress gradients are expected.

Role in the study

This mesh defines the spatial structure from which the COMSOL fields are exported for further preprocessing.

COMSOL Temperature Field



Thermal excitation

The heated source creates a localized high-temperature region inside the geological medium.

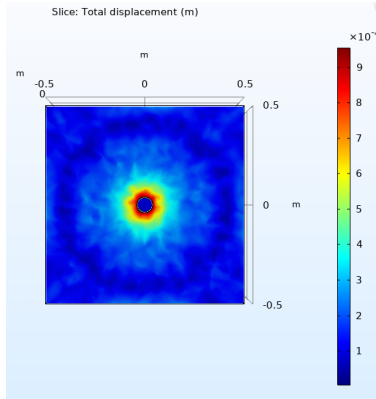
Temperature distribution

The maximum temperature is concentrated near the source, while the surrounding domain remains close to the initial temperature.

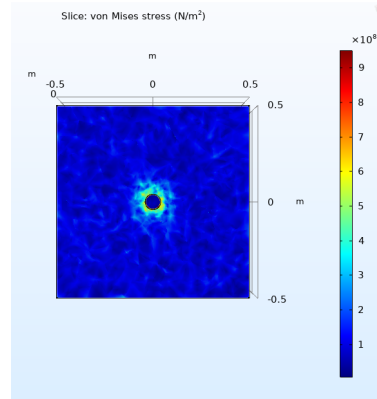
Dataset variable

Temperature is one of the main dynamic fields used for training and validating the neural models.

COMSOL Thermoelastic Response



Total displacement field



von Mises stress field

Local heating causes thermal expansion of the medium. As a result, displacement and stress are concentrated around the heated source, confirming the coupled thermoelastic behavior of the simulated process.

Data Representation and Preprocessing

- ▶ COMSOL simulation results are exported as CSV files for each field and time step.
- ▶ The exported data are aligned into a common node-based representation.
- ▶ Dynamic fields are assembled into a spatio-temporal tensor:

$$\mathbf{S} \in \mathbb{R}^{N_t \times N_p \times F_d}$$

- ▶ Static material and geometric features are stored separately.
- ▶ All feature channels are normalized before model training:

$$\tilde{x} = \frac{x - \mu}{\sigma + \varepsilon}$$

Here, N_t is the number of time steps, N_p is the number of spatial points, and F_d is the number of dynamic physical fields. The same processed dataset is then reused by coordinate-based, grid-based, and graph-based neural models.

Comparative Prototype Framework

One workflow for all model routes

The frontend, backend orchestrator, material catalog, and model services use one shared input–output contract for fair comparison.

- ▶ Same source–probe geometry for all routes
- ▶ Same effective geological material presets
- ▶ Same normalized response format: temperature, displacement, direction, travel time, and diagnostics

Shared Model Comparison Contract

Common interpretation

All routes are treated as surrogate modelling approaches for the same AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media task, not as complete numerical solvers.

Input

Material parameters, source–probe geometry, observation time, and processed field features.

Output

Estimated temperature, displacement components, travel time, and diagnostic response quantities.

Comparison basis

Same material presets, geometry contract, response format, and analytical travel-time sanity check.

Physics-Informed Neural Network (PINN)

Role

Physics-informed baseline in the comparison.

Input / output

Input: coordinates, time, and material parameters. Output: estimated temperature and in-plane displacement.

Why and limitation

Used because thermoelastic residuals can be included in training. It is still a baseline model, not an exact equation solver.

Model relation

$$f_{\theta}(x, y, t, E, \nu, \rho, \alpha, k, C_p) \rightarrow (T, u, v)$$

$$\mathcal{L} = \lambda_{\text{sup}} \mathcal{L}_{\text{sup}} + \lambda_{\text{vel}} \mathcal{L}_{\text{vel}} + \lambda_{\text{el}} \mathcal{L}_{\text{elastic}} + \lambda_{\text{th}} \mathcal{L}_{\text{thermal}}$$

x, y – coordinates [m]; t – time [s]; E – Young's modulus [Pa]; ν – Poisson's ratio [dimensionless]; ρ – density [kg/m^3]; α – thermal expansion [$1/\text{K}$]; k – conductivity [$\text{W}/(\text{m K})$]; C_p – heat capacity [$\text{J}/(\text{kg K})$]; T, u, v – predicted temperature [K] and displacements [m]. $\mathcal{L}$ and $\mathcal{L}_{\cdot}$ are normalized loss terms; $\lambda_{\cdot}$ are dimensionless loss weights.

PINN: Physics-Informed Baseline

Role in the comparison

The Physics-Informed Neural Network (PINN) is the route whose loss explicitly contains thermoelastic equation residuals.

- ▶ Candidate predictor for the shared task.
- ▶ Physics-informed baseline for the data-driven routes.
- ▶ Uses supervised data and governing-equation constraints.

Interpretation

PINN is not an exact numerical solver. It is a neural surrogate whose training objective encourages physically consistent predictions.

PINN Input–Output Mapping

Pointwise thermoelastic prediction

$$f_{\theta}(x, y, t, E, \nu, \rho, \alpha, k, C_p) \rightarrow (T, u, v)$$

- ▶ (x, y) : in-plane coordinate.
- ▶ t : observation time.
- ▶ $E, \nu, \rho, \alpha, k, C_p$: material parameters.
- ▶ T : predicted temperature.
- ▶ u, v : in-plane displacement components.
- ▶ Stress and strain are recovered from derivatives.

PINN Network Structure

Multilayer perceptron

$$h^{(\ell)} = \varphi \left(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)} \right)$$
$$h^{(0)} = (x, y, t, E, \nu, \rho, \alpha, k, C_p)$$

The implementation uses a smooth activation such as $\tanh$, because the PDE residuals require derivatives of the network output with respect to space and time.

θ collects all trainable weights and biases; automatic differentiation provides the derivatives used in the loss.

PINN Composite Loss

Training objective

$$\mathcal{L} = \lambda_{\text{sup}}\mathcal{L}_{\text{sup}} + \lambda_{\text{vel}}\mathcal{L}_{\text{vel}} + \lambda_{\text{el}}\mathcal{L}_{\text{elastic}} + \lambda_{\text{th}}\mathcal{L}_{\text{thermal}}$$

Data terms

- ▶ supervised error for (T, u, v) ;
- ▶ velocity consistency with exported COMSOL velocities.

Physics terms

- ▶ elastic-wave residual;
- ▶ coupled thermal residual.

PINN Thermoelastic Residuals

Momentum balance

$$\begin{aligned}\rho \partial_t^2 u - \nabla \cdot \sigma &= 0 \\ \sigma &= \lambda(\nabla \cdot u)I + 2\mu\varepsilon - \gamma\theta I\end{aligned}$$

Coupled heat equation

$$\rho C_p \partial_t T - k \nabla^2 T + \gamma T_0 \partial_t (\nabla \cdot u) = 0$$

$\theta = T - T_{\text{ref}}$ and $\gamma = 3K\alpha$ connect temperature, strain, and stress inside the residual terms.

PINN Data Use and Resulting Role

Training data

PINN uses the same COMSOL-derived 2D datasets: coordinates, time, material parameters, and ground-truth temperature and displacement.

Residuals are evaluated at collocation points using automatic differentiation.

Why it matters

Because governing-equation residuals are part of training, PINN is the physically grounded reference route in the comparison framework.

Its outputs should still be interpreted within the simplified 2D assumptions and available training-data coverage.

Fourier Neural Operator (FNO)

Role

Neural operator surrogate for structured two-dimensional fields.

Input

Grid-based material and response features, together with the shared source–probe and time query context.

Output

Estimated field-style response used to derive directional quantities.

Why and limitation

Included because Fourier operators can capture broad field patterns. In the current prototype, its output scale needs recalibration.

FNO: Field-to-Field Prediction Idea

Why FNO fits this task

Thermoelastic simulation output is not a single scalar. It is a set of spatial fields defined on a two-dimensional grid.

- ▶ The model learns an operator between function spaces.
- ▶ Inputs and outputs keep spatial structure.
- ▶ The target can include temperature and displacement fields.

Operator view

$$\mathcal{G}_\theta : \mathcal{A} \rightarrow \mathcal{U}, \quad \hat{s} = \mathcal{G}_\theta(a)$$

a – input field; $\hat{s}$ – predicted thermoelastic field; $\mathcal{A}$ and $\mathcal{U}$ – input and output field spaces; θ – trainable neural-operator parameters.

FNO Tensor Format for the 2D Setup

Input tensor

$$A \in \mathbb{R}^{B \times C_{in} \times N_x \times N_y}$$

Output tensor

$$\hat{S} \in \mathbb{R}^{B \times C_{out} \times N_x \times N_y}$$

- ▶ B is batch size.
- ▶ N_x, N_y are grid sizes.
- ▶ C_{in} is the number of input channels.
- ▶ C_{out} is controlled by `out_channels`.

For the three-channel thermoelastic setup:

$$C_{out} = 3, \quad \hat{s} = [\hat{T}, \hat{u}, \hat{v}].$$

FNO Architecture Used in the Prototype

Main processing chain

input $\rightarrow$ lifting $\rightarrow$ Fourier blocks $\rightarrow$ projection $\rightarrow$ output

Implementation parameters

- ▶ modes: retained Fourier modes
- ▶ num_fourier_layers: number of blocks
- ▶ mid_channels: latent capacity
- ▶ projection_channels: output projection size

Operator composition

$$\mathcal{G}_\theta = Q \circ \mathcal{B}_{L-1} \circ \cdots \circ \mathcal{B}_0 \circ P$$

P – lifting layer; $\mathcal{B}_l$ – Fourier block; Q – projection layer; L – number of Fourier blocks.

FNO Spectral Block

Block update

$$v_{l+1} = \sigma(\mathcal{S}_l(v_l) + \mathcal{C}_l(v_l))$$

The spectral branch captures global frequency-domain interactions; the local convolution branch captures neighboring spatial patterns.

Spectral convolution

$$\hat{v}_l = \mathcal{F}(v_l), \quad z_l = \mathcal{F}^{-1}(R_l \cdot \hat{v}_l)$$

$\mathcal{F}$ and $\mathcal{F}^{-1}$ are Fourier and inverse Fourier transforms; R_l is the trainable complex-valued spectral weight tensor.

Mode Truncation and Coordinate Encoding

Retained modes

$$\text{modes} = [K_1, K_2], \quad d = 2$$

Only selected low-frequency Fourier coefficients are transformed by learned weights. This controls capacity and computational cost.

Optional grid channels

$$\tilde{a}(x, y) = [a(x, y), x, y]$$

With `add_grid`, the model receives explicit normalized spatial coordinates. This helps distinguish source, boundary, and interior regions.

Training, Evaluation, and Current FNO Caveat

Supervised objective

$$\mathcal{L} = \lambda_{MSE} \mathcal{L}_{MSE} + \lambda_{rel} \mathcal{L}_{rel}$$

Channel normalization is computed on the training set and reused for validation, testing, and inference.

$$\tilde{s}_{i,c} = \frac{s_{i,c} - \mu_c}{\sigma_c}$$

Interpretation in this thesis

FNO is valuable as a field-based neural-operator route, but the current prototype still needs output-scale recalibration before it can be read as a physically reliable predictor.

Evaluation should use field-level plots, channel-wise errors, relative L^2 error, and travel-time sanity checks.

MeshGraphNet

Role

Graph-based surrogate model for mesh-aware prediction.

Input

Mesh nodes, graph connectivity, material features, and query context.

Output

Node-level or probe-level response estimates after graph message passing.

Why and limitation

Included because graph structure can represent local mesh interactions. The current route behaves mainly as a fixed-horizon predictor.

MeshGraphNet: Why a Graph Model

Reason for using MeshGraphNet

COMSOL finite-element data are naturally mesh-based. Nodes can have different neighbourhoods and nonuniform distances, unlike pixels on a regular image grid.

- ▶ Preserves unstructured mesh connectivity.
- ▶ Avoids forced interpolation to a Cartesian grid.
- ▶ Matches local thermal and mechanical interactions.

Graph representation

$$\mathcal{G} = (\mathcal{V}, \mathcal{E})$$

$\mathcal{V}$ – mesh nodes; $\mathcal{E}$ – local graph edges. Each node represents a spatial location in the simulated geological medium; each edge represents local physical neighbourhood connectivity.

Mesh and Edge Features

Geometric edge encoding

$$r_{ij} = p_j - p_i, \quad d_{ij} = \|r_{ij}\|_2$$
$$a_{ij} = \begin{bmatrix} r_{ij} \\ d_{ij} \end{bmatrix}$$

The edge attributes describe both direction and distance between neighbouring mesh nodes.

This matters because heat transfer, stress redistribution, and elastic wave response depend on local geometry, not only on node values.

p_i, p_j – node coordinates [m]; r_{ij} – relative vector [m]; d_{ij} – Euclidean edge length [m].

Node Features and Prediction Target

Node feature vector

$$x_i^t = \begin{bmatrix} p_i \\ m_i \\ s \\ q_i^t \end{bmatrix}$$

Coordinates, material parameters, scenario conditions, and current dynamic fields are concatenated before model input.

State transition

$$\hat{Y}^{t+\Delta t} = f_{\theta}(X^t, \mathcal{G})$$

$X^t \in \mathbb{R}^{N \times F_v}$ is the node feature matrix; $\hat{Y}^{t+\Delta t} \in \mathbb{R}^{N \times C}$ is the predicted future field matrix for all graph nodes.

Delta Prediction Mode

Absolute mode

$$\hat{q}_i^{t+\Delta t} = f_{\theta}(x_i^t, \mathcal{G})$$

Delta mode

$$\begin{aligned}\Delta \hat{q}_i^t &= f_{\theta}(x_i^t, \mathcal{G}) \\ \hat{q}_i^{t+\Delta t} &= q_i^t + \Delta \hat{q}_i^t\end{aligned}$$

Delta prediction is useful because neighbouring time-step changes are often smoother and smaller than full thermoelastic fields.

Rollout caveat

In autoregressive prediction, each predicted state becomes the next input. Small one-step errors can accumulate over long horizons.

Encoder–Processor–Decoder Structure

Model composition

$$f_{\theta} = D_{\theta_D} \circ P_{\theta_P} \circ E_{\theta_E}$$

Encoder

Maps raw node and edge features into latent vectors.

Processor

Applies repeated message passing over mesh connectivity.

Decoder

Converts final latent node states into predicted fields.

The repository implementation uses multilayer perceptrons, smooth SiLU activations, residual connections, and layer normalization inside the graph processing path.

Message Passing, Training, and Limits

Message passing step

$$\tilde{e}_{ij}^{k+1} = \phi_e^k([h_i^k, h_j^k, e_{ij}^k])$$

$$m_j^{k+1} = \sum_{i:(i,j) \in \mathcal{E}} e_{ij}^{k+1}$$

$$h_j^{k+1} = \phi_v^k([h_j^k, m_j^{k+1}])$$

Training objective

$$\mathcal{L} = \frac{1}{\sum_c w_c} \sum_{c=1}^C w_c \frac{1}{N} \sum_{i=1}^N (\hat{y}_{i,c} - y_{i,c})^2$$

MeshGraphNet is a supervised COMSOL surrogate. It preserves mesh topology and supports material conditioning, but physical equations are learned implicitly and must be checked through validation plots and metrics.

Transformer-Style Surrogate Model

Role

Attention-based baseline surrogate for fast query prediction.

Input

Tokens describing geometry, material parameters, state variables, and requested observation time.

Output

Fast estimates of temperature, displacement, travel time, and diagnostics.

Why and limitation

Included as a flexible low-latency route. Its physics priors are limited, so results require cautious interpretation.

Transformer: Attention-Based Surrogate

Role in the framework

The Transformer-style route is an attention-based surrogate for fast prediction under the shared input-output contract.

- ▶ Treats sampled points as tokens.
- ▶ Relates distant points through attention.
- ▶ Supports query-style prediction at requested locations.

Interpretation

This route is mainly data-driven. It can be fast and flexible, but its physical priors are weaker than those of the PINN route.

Transformer Tokenisation

Input token

$$a_i = [x_i, y_i, T_i^{(0)}, u_i^{(0)}, v_i^{(0)}, \dot{u}_i^{(0)}, \dot{v}_i^{(0)}, E_i, \nu_i, \rho_i, \alpha_i, k_i, C_{p,i}]$$

Query token

$$q_j = (x_q^{(j)}, y_q^{(j)})$$

Input tokens describe known mesh samples. Query tokens describe where the predicted fields are requested.

Scaled Dot-Product Attention

Attention operation

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^{\top}}{\sqrt{d_k}} \right) V$$

- ▶ Q stores query representations.
- ▶ K stores key representations.
- ▶ V stores value representations.

The score matrix connects every query with every input token. This gives the model a direct route for long-range interactions in the domain.

Transformer Encoder and Decoder

Encoder

Input tokens are projected to an embedding space and refined through multi-head self-attention and feed-forward layers.

$$H^{(\ell+1)} = \text{EncoderBlock}(H^{(\ell)})$$

Decoder

Query tokens attend to encoded input tokens through cross-attention.

$$Z^{(\ell+1)} = \text{DecoderBlock}(Z^{(\ell)}, H)$$

Transformer Output Head

Predicted fields at query points

$$\hat{y}_j = (\hat{T}_j, \hat{u}_j, \hat{v}_j)$$
$$\hat{y}_j = W_{\text{out}} Z_{j,:}^{(L_{\text{dec}})}$$

The model predicts temperature and two in-plane displacement components. Stress and strain can be derived later from displacement gradients or added as future output channels.

The baseline described in the source text uses an embedding dimension of $d_m = 128$ and multi-head attention with $h = 4$ heads.

Transformer Training Objective

Hybrid supervised objective

$$\mathcal{L} = \lambda_{\text{sup}}\mathcal{L}_{\text{sup}} + \lambda_{\text{vel}}\mathcal{L}_{\text{vel}} + \lambda_{\text{th}}\mathcal{L}_{\text{th}}$$

- ▶ supervised reconstruction of reference fields;
- ▶ velocity consistency for displacement evolution;
- ▶ thermal residual regularisation.

Token subsampling controls attention cost and encourages operation on different spatial samplings. Predictions still require careful validation against COMSOL-derived references.

Calculation Status

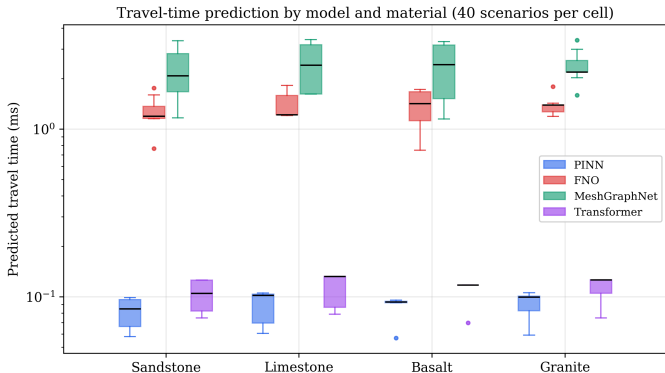
Calculation setup

1. **40 scenarios per material**, four materials: granite, sandstone, limestone, and basalt.
2. $40 \times 4 \times 4 = 640$ **model responses** evaluated under one shared contract.
3. **Zero fallback**: every plotted response is an active model output, not a mock.
4. **2D plane-strain only**: the operating envelope of all trained surrogates.

FNO caveat

FNO is included in the comparison, but its output scales are still miscalibrated. It should currently be treated as a diagnostic branch.

Travel-Time Predictions

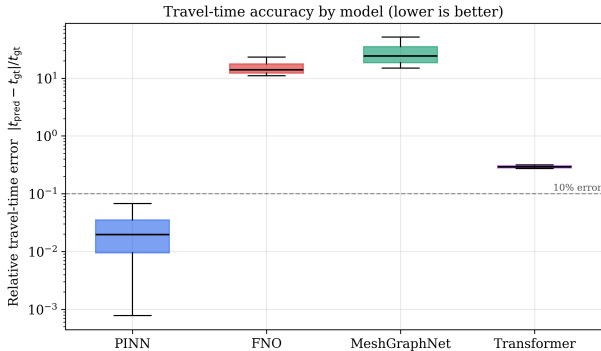


Reading the plot

- ▶ PINN ≈ 0.11 ms
- ▶ Transformer ≈ 0.14 ms
- ▶ FNO ≈ 1.29 ms
- ▶ MeshGraphNet ≈ 2.47 ms

Model-family differences are larger than the basalt–sandstone material difference in the current prototype.

Accuracy Against Analytical Reference



Analytical reference

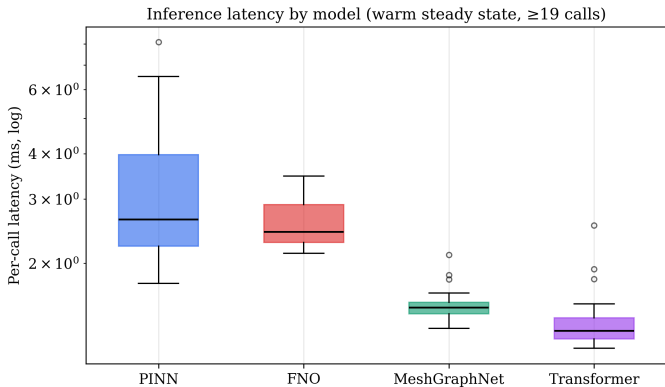
$$t_{\text{gt}} = \frac{r}{V_p}$$

t_{gt} – reference travel time [s]; r – source–probe distance [m]; V_p – compressional wave velocity [m/s].

Model	Median error	$\leq 10\%$
PINN	2.0%	100%
Transformer	29.2%	0%
FNO	1403.6%	0%
MeshGraphNet	> 100%	0%

PINN is the only route that remains inside the 10% band across the balanced sweep.

Inference Latency and Time Behavior



Median latency

- ▶ Transformer ≈ 1.3 ms
- ▶ MeshGraphNet ≈ 1.5 ms
- ▶ FNO ≈ 2.4 ms
- ▶ PINN ≈ 2.6 ms

PINN is also the only route that changes smoothly with requested observation time; the other routes currently behave as fixed-horizon predictors.

Threats to Validity

1. **Limited training data.** The COMSOL-derived pilot configuration is not a full calibrated geological database.
2. **Simplified 2D assumptions.** The study uses effective material parameters in a plane-strain setup.
3. **FNO normalization issue.** The current 2D adaptation is not yet physically reliable.
4. **Analytical reference is isotropic.** r/V_p is a useful sanity check, not a complete thermoelastic validation.

Conclusions

1. The thesis develops a physically grounded *comparative framework* for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media.
2. PINN is the strongest *physics-informed baseline* in the current comparison and shows the best agreement with the analytical travel-time reference.
3. Transformer is the fastest route, while MeshGraphNet remains a useful mesh-aware surrogate candidate.
4. FNO should currently be treated as a diagnostic branch until output normalization is corrected.

Future Work

1. Retrain and recalibrate the 2D FNO route.
2. Add time-conditioned rollout behavior to MeshGraphNet.
3. Extend the material catalog with more complete thermoelastic parameters and validation data.
4. Compare against additional analytical or numerical baselines.

Final message

The prototype estimates directional characteristics of thermoelastic wave propagation within the simplified assumptions of the study.

Thank You

Questions?